

Object-Oriented Programming

Luisa Daniela Correa Torres - 20242020048

Juan Pablo Rodríguez - 20251020072

Bryan Martinez Carreño - 20242020257

Workshop No. 2 — Object-Oriented Design

Universidad Distrital Francisco José de Caldas

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1 Project Definition

1.1 Project Name

Prestige Eventos & Recepciones

1.2 Description

This project is for people planning a big party, like a wedding or a large birthday. Right now, planning something like this is hard because you have to call many different people (for the place, the food, and the decorations) and try to put all the pieces together yourself. This system makes it easy. It's a single website where you can see what dates are available, choose a party theme, pick the food you want, and add other services. You get to see the total price right away and book everything at once. The main goal is to take the stress out of planning a big event, making it a simple and clear process for everyone.

1.3 Main Features

The system guides you through planning your entire event in one place. You can start by choosing a fun theme for your party, like a classic wedding or a tropical birthday, which suggests matching decorations. Then, you'll pick the perfect food and drinks for your guests, from a formal dinner to a casual buffet. Finally, you can check the availability of your favorite venue, lock in your ideal date, and tell us how many people are coming to get everything officially booked. To make it easy, you can create a personal account to save your ideas and manage all your party details securely.

2 Requirements

2.1 Functional Requirements

- User registration and authentication: Customers can create an account and log in to manage their reservations.
- Date calendar: The user can check availability on an interactive calendar.
- Event booking: The client selects the date, type of party, number of guests, and additional services.
- Modify: The user must be able to cancel or modify their reservation within two weeks.
- Support: The user should be able to communicate with us in the way that best suits them.
- Service Packages: The system must allow users to view and select predefined service packages that combine venue, theme, and catering options.
- Online Payments: The user must be able to securely complete online payments for their reservations and receive immediate payment confirmation.

- Automatic Notifications: The system must automatically generate and send email confirmations for both completed bookings and any subsequent modifications made to the reservation.

2.2 Non-Functional Requirements

- Ease of Use: The design must be simple and clear so that anyone, even without experience, can easily book their event.
- Mobile-Friendly: The website must work perfectly and look good on any device, like smartphones and tablets.
- Security: User data and payment information must be completely protected with modern security measures.
- Scalability: The system must work smoothly even when many people are using it at the same time without slowing down or crashing.
- Security: User data and payment information must be completely protected with modern security measures.
- Availability: The system must be working and available to users 24/7, especially on weekends when most people plan events.

3 User Stories

Of course. Here are the user stories, each in its own LaTeX table, formatted correctly with the "so that" clause, acceptance criteria, priority, and effort estimation.

User Story 1

Title	Choose the Best Party Package
As a	client,
I want to	see and compare different pre-defined party packages (that include a theme, food, and decorations),
so that	I can easily choose the one that best fits my needs and vision without having to plan every detail myself.
Priority	High
Effort	5
Acceptance Criteria	• The main page must show a gallery of available party packages. • Each package must display a clear title, a main image, a short description, and a starting price. • I must be able to click on a package to see a detailed page with a full list of what's included (venue, food options, decoration style, etc.).

User Story 2

Title	Check Availability on a Calendar
As a	client,
I want to	see an interactive calendar that shows which dates are available for my chosen venue,
so that	I can quickly find and select a date that works for me and my guests.
Priority	High
Effort	3
Acceptance Criteria	• The calendar must be visually clear, with available dates marked differently from booked dates. • Clicking on an available date should allow me to proceed with the booking for that date. • If a date is unavailable, it should be clearly marked and not selectable.

User Story 3

Title	Modify or Cancel a Reservation
As a	customer,
I want to	cancel or make changes to my reservation (like the number of guests or food menu),
so that	I have the flexibility to adjust my plans if my needs change.
Priority	Medium
Effort	5
Acceptance Criteria	• From 'My Account,' I must see a list of my active reservations with an 'Edit' and 'Cancel' button. • I can only cancel or change my reservation up to 14 days before the event date. • After any change, the system must show me an updated summary and a new total price. • I must receive an email confirmation after any modification or cancellation.

User Story 4

Title	Create a Personal Account
As a	customer,
I want to	create a personal account using my email and a password,
so that	I can save my preferences, see my booking history, and make future reservations faster.
Priority	High
Effort	3
Acceptance Criteria	• The registration form must only ask for my full name, email, and a secure password. • I must receive a verification email to activate my account. • Once logged in, I must have a "My Account" page where I can see my profile and past bookings.

User Story 5

Title	Contact the Organizers Easily
As a	customer,
I want to	have multiple, clear ways to contact customer support (like a contact form and a phone number),
so that	I can get help in the way that is most convenient for me if I have a question or problem.
Priority	Low
Effort	2
Acceptance Criteria	• The website's footer and contact page must display a customer service phone number and email address. • A contact form must be available that sends my message directly to the support team. • After submitting the form, I must see an on-screen confirmation that my message was sent.

4 CRC Cards

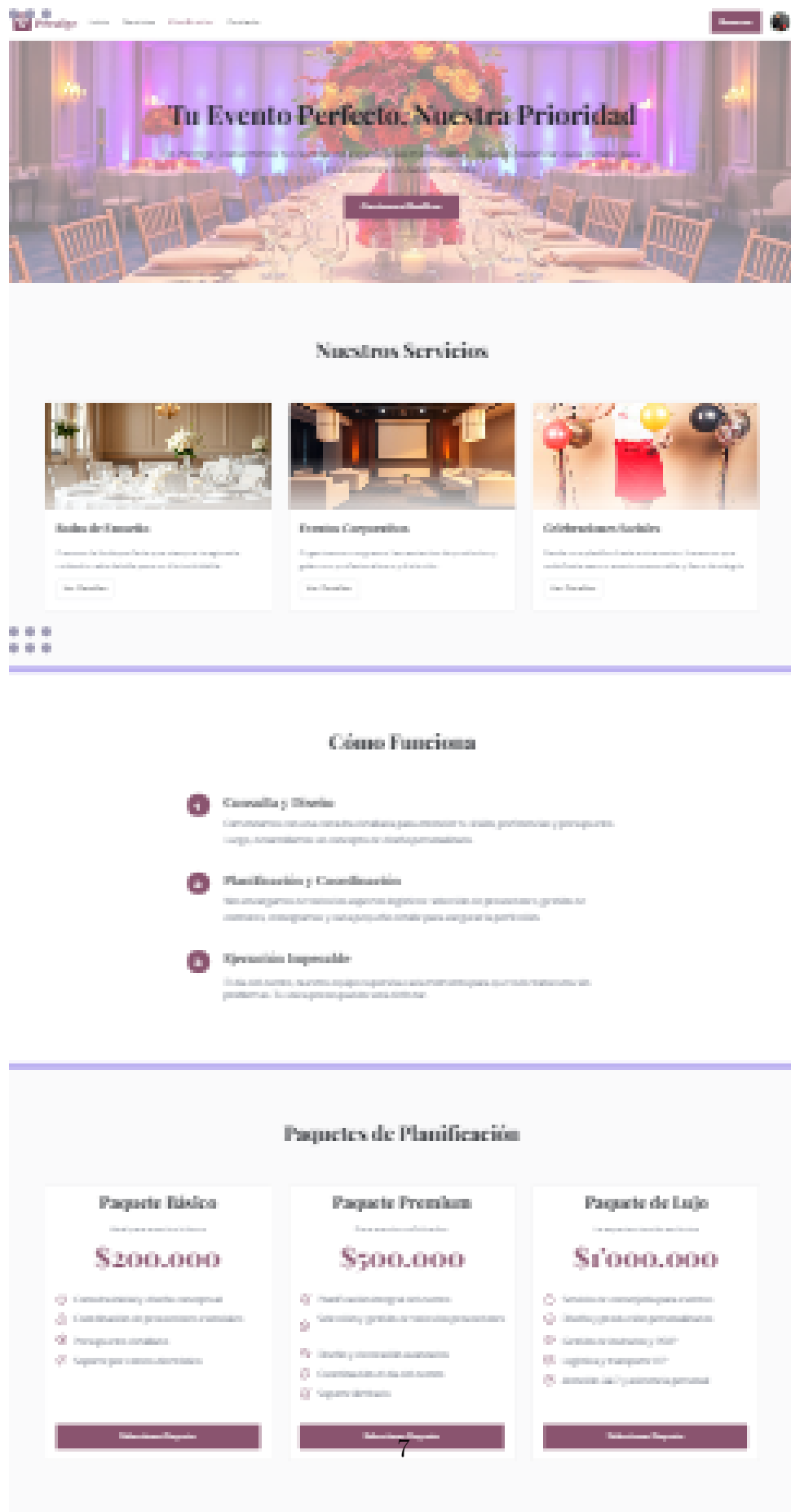
Class: Venue	Class: User
Responsibilities: • Save place info • Check dates • Verify capacity • Manage calendar	Responsibilities: • Create accounts • Save user data • Manage preferences
Collaborators: • Reservation • Calendar	Collaborators: • Reservation • Payment

Class: Calendar	Class: Payment
Responsibilities: • Show dates • Block dates • Date selection	Responsibilities: • Process payments • Payment methods • Generate receipts • Handle refunds
Collaborators: • Venue • Reservation	Collaborators: • Reservation • User

Class: Service	Class: Reservation
Responsibilities: • Manage services • Save prices • Check availability • Calculate cost	Responsibilities: • Create reservations • Calculate total • Change status • Manage dates
Collaborators: • Reservation	Collaborators: • User • Venue • Service • Payment

5 Mockups

User Interface Designs



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Desde exquisitos platillos para eventos íntimos, hasta banquetes impresionantes, **Prestige Catering** ofrece opciones para cada ocasión. Descubren nuestra variedad e historia y personaliza tu experiencia gastronómica.

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Ensalada Caprese Fresca

Una deliciosa mezcla de tomates cherry maduros, queso mozzarella fresco y hojas de albahaca, adornada con una vinagreta balsámica. Perfecta como acompañamiento o entrante.

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COP \$25.000

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Filete Mignon con Salsa de Champiñones

Deliciosos filetes de bife de res seleccionados con precisión, servidos con una salsa cremosa de champiñones. Acompañados de espárragos frescos y puré de papa.

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Tarta de Limón con Merengue Suizo

Una exquisita tarta con una base de galleta crujiente, rellena de mermelada de limón y coronada con un merengue suizo cremoso. Ideal para cualquier ocasión especial.

[Ver receta](#) [Solicita](#)

COP \$30.000

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Paella Valenciana Tradicional

Auténtica paella valenciana con pollo, conejo, judías verdes y garbanos, sazonada con azafrán para lograr el color y aroma perfectos. Un clásico que transporta a todos los comensales.

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Tabla de Quesos Artesanales

Una selección variada de quesos artesanales de diferentes regiones, acompañada de frutos secos, miel, mermeladas y pan artesanal. Perfecta para compartir y disfrutar.

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Brochetas de Camarones al Ajillo

Deliciosas brochetas de camarones al ajillo, cocinados en aceite de oliva y especias, acompañados de una deliciosa salsa de tomate y cebolla.

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COP \$18.000

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Lougou Casero Boleto

Un aperitivo casero hecho a mano, con una base de pan artesanal relleno de champiñones y queso, acompañado de una deliciosa salsa de tomate y cebolla.

[Ver receta](#) [Solicita](#)

COP \$10.000

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Mesa de Postres Variados

Una selección personalizada de postres que incluye macarons, tartas de temporada, pasteles de chocolate y otros dulces artesanales. Ideal para cualquier ocasión especial.

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Bebidas Refrescantes Naturales

Una selección variada de bebidas naturales, sin azúcar, hechas con frutas frescas y agua mineral. Perfectas para acompañar cualquier comida y mantenerse hidratados.

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¿Listo para planificar tu evento?

Contáctanos hoy para diseñar un menú perfecto que se adapte a tus invitados. Nuestros equipos están listos para ayudarte en cada detalle.

[Contáctanos Ahora](#)



Contáctenos

Nos encantaría saber de usted. Envíenos un mensaje y nos pondremos en contacto pronto.

Nombre

Correo Electrónico

Teléfono

Mensaje

Enviar Mensaje

Nuestra Ubicación

Descubramos en el corazón de la ciudad para sus eventos.



Detalles de Contacto

+52 1 234 5678

info@prestigeevents.com

Lunes a Viernes: 9:00 AM - 6:00 PM



Prestige Events: su espacio para experiencias inolvidables, creando momentos mágicos.



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Contáctenos

Nos encantaría saber de usted. Envíenos un mensaje y nos pondremos en contacto pronto.

Nombre

Correo Electrónico

Teléfono

Mensaje

Enviar Mensaje

Nuestra Ubicación

Descubramos en el corazón de la ciudad para sus eventos.





Solicita tu Cotización Personalizada

Planifica el evento de tus sueños con facilidad.
 Completa nuestro formulario para recibir una
 cotización detallada y adaptada a tus necesidades.
 Podemos ayudarte a hacer realidad tu visión.



Detalles del Evento

Completa estos campos para que podamos ayudarte.

Tipo de Evento

Fecha del Evento

Forma de Pago

Temas y Preferencias

Elige los temas que te gustan y añade cualquier preferencia especial.

Temas seleccionados:



Rústico



Moderno



Glamorosa



Bohemio



Clásico

Preferencias especiales:

Información de Contacto

Completa estos campos para que podamos contactarte.

Nombre Completo

Correo Electrónico

Número de Teléfono

Resumen de tu Solicitud

Resumen de los datos que has introducido en el formulario.

Detalles del Evento

Tipo	No especificado
Fecha	No especificado
Forma de Pago	No especificado

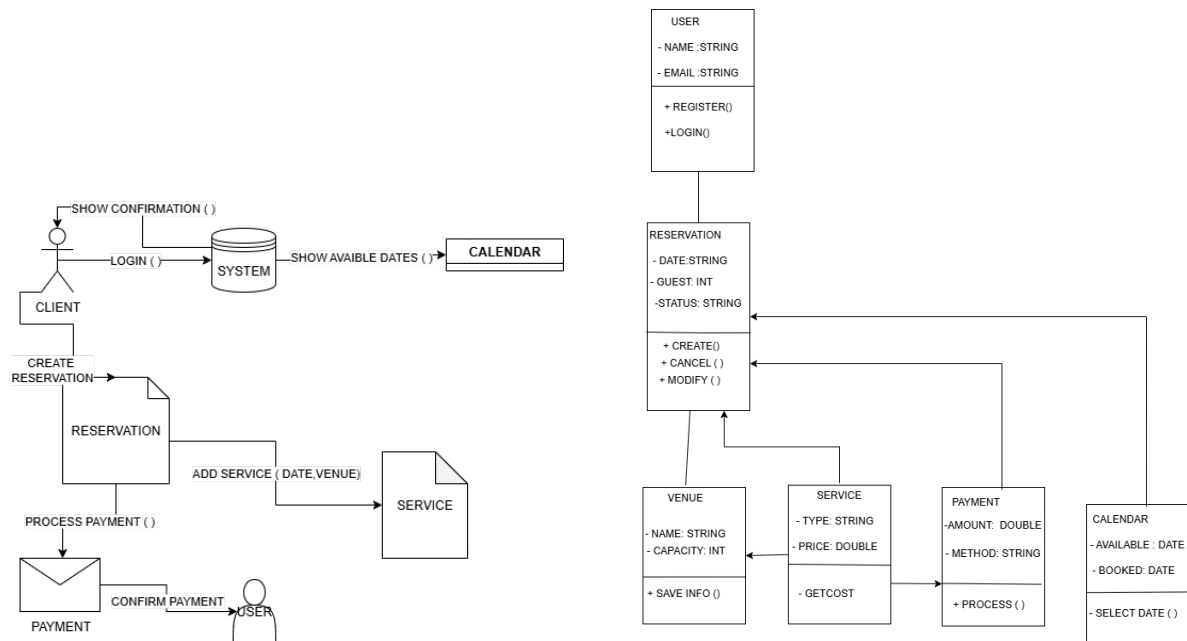
Preferencias de Tema

Temas seleccionados:	Ninguno
Preferencias especiales:	Ninguna

Datos de Contacto

Nombre	No especificado
Email	No especificado
Teléfono	No especificado

6 UML DIAGRAMS



7 Java Implementation

7.1 Base Classes according to UML

Client.java - User interacting with the system

```

public class Client {
    private String name, email, password;
    public Client(String name, String email, String password) {
        this.name = name; this.email = email; this.password = password;
    }
    public boolean login(String email, String password) {
        return this.email.equals(email) && this.password.equals(password);
    }
    public void showConfirmation(String message) {
        System.out.println("Confirmation: " + message);
    }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
}
  
```

UML: `login()`, `showConfirmation()` methods. Private attributes encapsulation.

SystemApp.java - Main application

```

public class SystemApp {
    private Calendar calendar;
    public SystemApp(Calendar calendar) { this.calendar = calendar; }
    public void showAvailableDates() { calendar.displayAvailableDates(); }
}
  
```

UML: SHOW AVAILABLE DATES() between SYSTEM-CALENDAR.

Calendar.java - Date management

```
import java.util.*;
public class Calendar {
    private List<String> availableDates;
    public Calendar() {
        availableDates = new ArrayList<>();
        availableDates.add("2025-11-20");
        availableDates.add("2025-12-05");
    }
    public void displayAvailableDates() {
        System.out.println("Available Dates:");
        for (String date : availableDates) { System.out.println(date); }
    }
}
```

UML: Shows available dates for reservations.

Service.java - Event services

```
public class Service {
    private String name, venue;
    private double price;
    public Service(String name, double price, String venue) {
        this.name = name; this.price = price; this.venue = venue;
    }
    public void addService(String date, String venue) {
        System.out.println("Service '" + name +
            "' added for " + date + " at " + venue);
    }
    public double getPrice() { return price; }
}
```

UML: SERVICE connected via "ADD SERVICE (DATE, VENUE)".

Reservation.java - Event reservations

```
import java.util.*;
public class Reservation {
    private Client client;
    private List<Service> services;
    private String date;
    public Reservation(Client client, String date) {
        this.client = client; this.date = date;
        this.services = new ArrayList<>();
    }
    public void createReservation() {
        System.out.println("Reservation created for " +
            client.getName() + " on " + date);
    }
}
```

```

        public void addService(Service service) {
            services.add(service);
            System.out.println("Service added: " + service);
        }
    }
}

```

UML: CREATE RESERVATION process linked with SERVICE.

Payment.java - Payment processing

```

public class Payment {
    private double amount;
    private boolean isProcessed;
    public Payment(double amount) {
        this.amount = amount; this.isProcessed = false;
    }
    public void processPayment() {
        this.isProcessed = true;
        System.out.println("Payment of $" + amount + " processed");
    }
    public void confirmPayment(Client client) {
        if (isProcessed) client.showConfirmation("Payment confirmed!");
        else System.out.println("Payment not processed yet");
    }
}

```

UML: PROCESS PAYMENT(), CONFIRM PAYMENT() methods.

7.2 Integration Example

Main.java - System integration

```

public class Main {
    public static void main(String[] args) {
        Calendar calendar = new Calendar();
        SystemApp system = new SystemApp(calendar);
        Client client = new Client("Maria", "maria@example.com", "1234");

        if (client.login("maria@example.com", "1234")) {
            system.showAvailableDates();
            Reservation reservation = new Reservation(client, "2025-11-20");
            reservation.createReservation();
            Service service = new Service("Catering", 500.0, "Main Hall");
            service.addService("2025-11-20", "Main Hall");
            Payment payment = new Payment(500.0);
            payment.processPayment();
            payment.confirmPayment(client);
        }
    }
}

```

7.3 REFLECTION

We realized that CRC cards are extremely important when developing any project. Without the CRC cards, we would not have been able to create the UML diagrams, and without the diagrams, programming would have been much more complicated since we wouldn't have had a clear foundation of what we wanted to build. We also redesigned the mockup to achieve a more visually appealing layout while keeping the main idea we had in Workshop 1, also took into account existing websites similar to the one we planned, using them as references to guide our design choices and to evaluate how visually appealing and user-friendly they are. for example **eventos bella suiza,cocteles y eventos,and comun y silvestre.**